ACTIVITY 9
Relays and Contactors, Push Buttons…

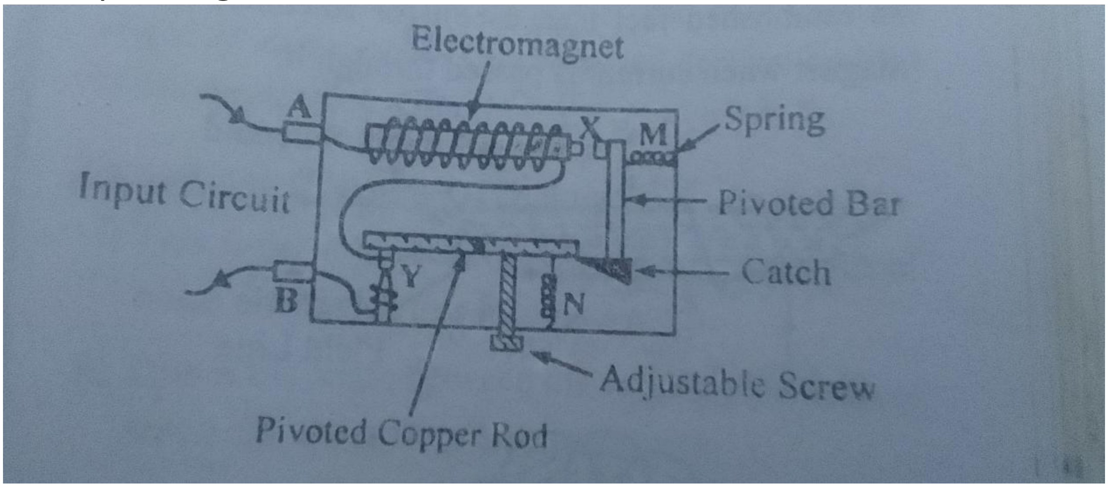
Circuit breaker:

A circuit breaker is an electrical safety device that automatically interrupts electrical flow in a circuit when it detects a fault, such as short circuit or overload.

It is designed to protect electrical systems from damage caused by excessive current and to prevent electrical fires.

The ability of the breaker to cut off current flow automatically is called tripping.

A simple diagram of the circuit breaker is as shown below.



Mode of operation of the circuit breaker:

Normal operation: in normal operation, the contacts in the breaker are closed, allowing current to flow through the circuit.

Overload: when an overload occurs, the current through the breaker increases. The solenoid in the breaker senses the increase in current and produces a magnetic field, which pulls on the contacts. The contacts open and interrupts the flow of current.

Short circuit: In a short circuit, the breaker increases to an extremely high level. The magnetic field produced by the solenoid becomes strong to instantly pull the contacts apart, opening the circuit and stopping the flow of current.

Types of circuit breakers:
Single pole: breaks/controls one circuit
Two pole: breaks/controls two circuits
Three pole: breaks/controls three circuits
Four pole: breaks or controls four circuits
Miniature circuit breaker:

Relay:
A relay is an electrical component used that uses an electromagnetic coil to control one or more switches.
It is used to switch on or off multiple circuits by sensing the changes in the electrical input signal or signals.
A relay is an electromechanical component that is designed to switch high voltage and high current loads to with a low voltage signal. In other words, a relay itself does not provide power, but it can be used to control the power to low voltage machines.
For example, if you have a low voltage machine that requires 12V DC power to operate, you could use a relay to switch on and off the power to that machine using a low voltage signal, such as 5V DC. When the relay is energized by the low voltage signal, it completes a circuit that allows power to flow to the machine. When the relay is de-energized, the circuit is broken and the power to the machine is turned off.
So, while a relay itself does not provide power to low voltage machines, it can be used as a switching mechanism to control the power to those machines.

Contactors:
A contactor is an electrical component that is designed to switch high voltage and high current loads on and off. It works by using an electromagnetic coil to activate the switch contacts, which then allows current to flow through the circuit.

The main components of a contactor include the electromagnetic coil, armature, switch contacts, contact springs, and enclosure.
Contactors are typically classified according to their current and voltage ratings, and there are many different types of contactors available to suit different applications. For example, there are three-phase contactors that are designed to switch three-phase electrical circuits, and motor control contactors that are specifically designed to control electric motors.

Push button:
The push button is a device used to activate or deactivate circuits.
A push button is a type of switch that is designed to be activated by a physical push or press of a button. It is commonly used in various applications such as control panels, machinery, and electronic devices.
Types: The start button is used to initiate the operation of the device. When the button is pressed, it sends a signal to the control circuit, which then switches on the power to the motor or device.
The stop button is used to halt the operation of the device. When the button is pressed, it sends a signal to the control circuit, which then switches off the power to the motor or device.
Both buttons are typically momentary switches, meaning that they only remain in the on-state while the button is being pressed. Once released, they return to the off-state.

Refrigeration:
Refrigeration is the process of removing heat from an enclosed space or substance, typically for the purpose of preserving or cooling a product or environment. Refrigeration is used in many applications, such as food storage and preservation, air conditioning, and industrial processes.

The basic principle of refrigeration is the transfer of heat from a low-temperature area to a high-temperature area using a refrigerant, which is a fluid that undergoes a cycle of compression, condensation, expansion, and evaporation. The refrigerant is circulated through a closed loop system that includes a compressor, condenser, expansion valve, and evaporator.

The compressor is responsible for compressing the refrigerant gas and pumping it into the condenser.

The condenser is where the refrigerant releases heat and condenses into a liquid.

The liquid refrigerant then flows through **the expansion valve,** which reduces its pressure and temperature.

As the refrigerant passes through **the evaporator**, it absorbs heat from the surrounding environment and evaporates back into a gas. This process continues in a cycle, with the refrigerant flowing back to the compressor to begin the process again.

A filter device is an important component in a refrigeration closed loop system as it helps to remove contaminants from the refrigerant.

Refrigerants used in refrigeration:

1. R12
2. R-134A
3. R22
4. R600A